# VALIDATION REPORT
# OPERATIONAL QUALIFICATION
# STATISTICAL PROCESS CONTROL MODULE

## JR Validated Environment — Statistical Process Control Module v1.0

| Field | Value |
|---|---|
| Document Number | JR-VR-SPC-001 |
| Title | Validation Report — Operational Qualification, Statistical Process Control Module |
| Validation Plan | JR-VP-SPC-001 v1.0 |
| System | JR Validated Environment — Statistical Process Control Module |
| Module Version | 1.0 |
| Document Version | 1.0 |
| Execution Date | 2026-03-31 |
| OQ Result | PASS — 71 / 71 tests passed, 0 failures, 0 deviations |
| Author | Joep Rous |
| Reviewer | [Name] |
| Approver | [Name] |

## Version History

| Version | Date | Author | Description |
|---|---|---|---|
| 1.0 | 2026-03-31 | Joep Rous | Initial OQ execution report. All 71 test cases passed. No deviations. |
| 1.1 | 2026-03-31 | Joep Rous | Report regenerated from evidence file. 71 / 71 tests passed. |

## Approval Signatures

| Role | Name | Signature | Date |
|---|---|---|---|
| Author | Joep Rous | | 2026-03-31 |
| Reviewer | [Name] | | |
| Approver (QA) | [Name] | | |

# 1. Purpose

This document reports the execution and outcome of the Operational Qualification (OQ) test suite for the JR Validated Environment Statistical Process Control Module, version 1.0. It provides objective evidence that all SPC scripts perform as specified in the OQ Validation Plan JR-VP-SPC-001 v1.0 under the defined test conditions.

This report records the test environment, execution results for all 71 test cases, and the formal OQ conclusion. Results are derived directly from the evidence file generated by admin_spc_oq.

# 2. Scope

## 2.1 In Scope

- All 5 R scripts in repos/spc/R/ (jrc_spc_imr, jrc_spc_xbar_r, jrc_spc_xbar_s, jrc_spc_p, jrc_spc_c)
- Shewhart control chart computations with 8 Western Electric rules
- PNG output to ~/Downloads/
- User-specified control limit override (--ucl / --lcl flags)
- Correct rejection of invalid inputs with informative error messages
- Bypass-protection verification (RENV_PATHS_ROOT check)
- Numeric correctness: control limits verified against independently computed reference values using Shewhart constants (d2=1.128, D4=3.267)
- OOC observation flagging — flagged subgroup or individual identified with rule number
- IN CONTROL / OUT OF CONTROL verdict for stable and out-of-control datasets

## 2.2 Out of Scope

- Infrastructure scripts in bin/ and admin/ (covered by JR-IQ-001)
- Performance qualification (PQ) with real production data
- Validation of R interpreter and third-party packages (ggplot2, qcc)
- Statistical validation of the underlying control chart theory — method references are cited in JR-VP-SPC-001 Section 3

# 3. Reference Documents

| Document ID | Title | Location |
|---|---|---|
| JR-VP-SPC-001 v1.0 | Validation Plan — OQ, Statistical Process Control Module | repos/spc/docs/spc_validation_plan.pdf |
| JR-VP-002 v1.0 | Validation Plan — OQ, Community Script Suite v1.1.0 | docs/oq_validation_plan.docx |
| JR-VP-001 v1.0 | Validation Plan — Installation Qualification | docs/ |
| JR-IQ-001 | IQ Execution Evidence | docs/IQ_validation_20260311_205146.txt |
| OQ Evidence | SPC OQ Execution Evidence (pytest output + header) | spc_oq_execution_20260331T194057.txt |

# 4. Test Environment

| Component | Details |
|---|---|
| Execution date / time | 2026-03-31T19:40:57 |
| Host | Joeps-MacBook-Air.local |
| Operating System | [not found] |
| R Version | 4.5.3 |
| Python Version | 3.11.9 |
| pytest Version | pytest 8.3.4 |
| Test runner | repos/spc/admin_spc_oq |
| Test directory | repos/spc/oq/ |
| Test data | repos/spc/oq/data/ (12 CSV fixtures) |
| Execution command | bash repos/spc/admin_spc_oq |
| OQ evidence file | spc_oq_execution_20260331T194057.txt |

## 5. Test Data

All test data CSV files reside in repos/spc/oq/data/. Each file contains two columns: id and value (or id, subgroup, value for subgroup charts). Numeric reference values were computed independently and are used as acceptance criteria for numeric correctness test cases.

| File | Description | Key Properties |
|---|---|---|
| imr_stable.csv | I-MR stable dataset | n=25 individuals; X-bar=10.0668, MR-bar=0.32125; no WE violations |
| imr_ooc.csv | I-MR out-of-control dataset | imr_stable with obs 13 = 11.50 (Rule 1 violation) |
| imr_missing_col.csv | I-MR missing column fixture | CSV with no 'value' column — triggers error handling |
| imr_one_obs.csv | I-MR too few observations fixture | CSV with only 1 row — triggers too-few-observations error |
| xbar_r_stable.csv | X-bar & R stable dataset | 20 subgroups × n=5; grand X-bar=50.165; no WE violations |
| xbar_r_ooc.csv | X-bar & R out-of-control dataset | xbar_r_stable with sg09 mean≈53 (Rule 1 violation) |
| xbar_s_stable.csv | X-bar & S stable dataset | 20 subgroups × n=8; grand X-bar=100.060; no WE violations |
| xbar_s_ooc.csv | X-bar & S out-of-control dataset | xbar_s_stable with sg09 mean≈102 (Rule 1 violation) |
| p_stable.csv | P-chart stable dataset | 25 subgroups n=100 each; p-bar=0.04160; no WE violations |
| p_ooc.csv | P-chart out-of-control dataset | p_stable with subgroup 13 = 15 defectives (Rule 1 violation) |
| c_stable.csv | C-chart stable dataset | 25 subgroups of defect counts; c-bar=4.960; no WE violations |
| c_ooc.csv | C-chart out-of-control dataset | c_stable with subgroup 12 = 15 defects (Rule 1 violation) |

## 6. Test Results

All 71 test cases were executed on 2026-03-31. Results are derived from the OQ evidence file and reproduced below. The evidence file contains the complete pytest output including individual test IDs, pass/fail status, and timing.

### 6.1 Summary

| Script | Tests Executed | Result |
|---|---|---|

| | | |
|---|---|---|
| jrc_spc_imr | 15 | 15/15 PASS |
| jrc_spc_xbar_r | 16 | 16/16 PASS |
| jrc_spc_xbar_s | 15 | 15/15 PASS |
| jrc_spc_p | 13 | 13/13 PASS |
| jrc_spc_c | 12 | 12/12 PASS |
| TOTAL | 71 | 71/71 PASS |

## 6.2 Evidence File

| Item | Details |
|---|---|
| Filename | spc_oq_execution_20260331T194057.txt |
| Timestamp | 2026-03-31T19:40:57 |
| Host | Joeps-MacBook-Air.local |
| Full path | spc_oq_execution_20260331T194057.txt |

## 6.3 jrc_spc_imr — 15/15 PASS

Script: jrc_spc_imr.R — I-MR (Individuals & Moving Range) chart. Test data: imr_stable.csv (stable), imr_ooc.csv (out-of-control).

| TC ID | Description | Expected | Result |
|---|---|---|---|
| TC-SPC-IMR-001 | Stable dataset — exit 0, control limit section and verdict present | Exit 0, control limits, stability section, verdict present | **PASS** |
| TC-SPC-IMR-002 | Stable data — IN CONTROL verdict | Exit 0, 'IN CONTROL' or 'STABLE' in output | **PASS** |
| TC-SPC-IMR-003 | OOC dataset — exit 0, OUT OF CONTROL in output | Exit 0, 'OUT OF CONTROL' or 'SIGNALS' in output | **PASS** |
| TC-SPC-IMR-004 | OOC dataset — observation 13 flagged as Rule 1 violation | Exit 0, '13' and '[1]' or 'Rule 1' in output | **PASS** |
| TC-SPC-IMR-005 | --ucl / --lcl flags accepted, user-specified limits reported | Exit 0, user-specified UCL mentioned | **PASS** |
| TC-SPC-IMR-006 | PNG written to ~/Downloads/ | PNG present, recent mtime | **PASS** |
| TC-SPC-IMR-007 | No arguments — non-zero exit, usage message | Exit ≠ 0, 'Usage' in output | **PASS** |
| TC-SPC-IMR-008 | File not found — non-zero exit | Exit ≠ 0, 'not found' in output | **PASS** |
| TC-SPC-IMR-009 | Missing 'value' column — non-zero exit, column named | Exit ≠ 0, 'value' in output | **PASS** |
| TC-SPC-IMR-010 | Too few observations (1 row) — non-zero exit | Exit ≠ 0 | **PASS** |
| TC-SPC-IMR-011 | Bypass protection — direct Rscript call fails | Exit ≠ 0, 'RENV_PATHS_ROOT' in output | **PASS** |
| TC-SPC-IMR-012 | X-bar = 10.0668 ± 0.0001 (imr_stable.csv) | X-bar: 10.0668 ± 0.0001 | **PASS** |
| TC-SPC-IMR-013 | UCL_I = 10.9212 ± 0.001 (imr_stable.csv) | UCL: 10.9212 ± 0.001 | **PASS** |
| TC-SPC-IMR-014 | LCL_I = 9.2124 ± 0.001 (imr_stable.csv) | LCL: 9.2124 ± 0.001 | **PASS** |
| TC-SPC-IMR-015 | UCL_MR = 1.0495 ± 0.001 (imr_stable.csv) | UCL_MR: 1.0495 ± 0.001 | **PASS** |

## 6.4 jrc_spc_xbar_r — 16/16 PASS

Script: jrc_spc_xbar_r.R — X-bar & R (X-bar and Range) chart. Test data: xbar_r_stable.csv (stable), xbar_r_ooc.csv (out-of-control).

| TC ID | Description | Expected | Result |
|---|---|---|---|
| TC-SPC-XBR-001 | Stable dataset — exit 0, X-bar and R sections and verdict present | Exit 0, X-bar, Range sections, verdict | **PASS** |
| TC-SPC-XBR-002 | Stable data — IN CONTROL verdict | Exit 0, 'IN CONTROL' or 'STABLE' | **PASS** |
| TC-SPC-XBR-003 | OOC dataset — exit 0, OUT OF CONTROL | Exit 0, 'OUT OF CONTROL' or 'SIGNALS' | **PASS** |
| TC-SPC-XBR-004 | OOC dataset — subgroup sg09 flagged as violation | Exit 0, 'sg09' in output | **PASS** |
| TC-SPC-XBR-005 | --ucl / --lcl flags accepted | Exit 0, user-specified UCL mentioned | **PASS** |
| TC-SPC-XBR-006 | PNG written to ~/Downloads/ | PNG present | **PASS** |
| TC-SPC-XBR-007 | No arguments — non-zero exit, usage message | Exit ≠ 0, 'Usage' | **PASS** |
| TC-SPC-XBR-008 | File not found — non-zero exit | Exit ≠ 0 | **PASS** |
| TC-SPC-XBR-009 | Missing column — non-zero exit | Exit ≠ 0 | **PASS** |
| TC-SPC-XBR-010 | Unbalanced subgroups — non-zero exit | Exit ≠ 0 | **PASS** |
| TC-SPC-XBR-011 | Subgroup size > 10 — non-zero exit, guidance message | Exit ≠ 0 | **PASS** |
| TC-SPC-XBR-012 | Bypass protection | Exit ≠ 0, 'RENV_PATHS_ROOT' | **PASS** |
| TC-SPC-XBR-013 | Grand X-bar = 50.165 ± 0.001 (xbar_r_stable.csv) | Grand X-bar: 50.165 ± 0.001 | **PASS** |
| TC-SPC-XBR-014 | UCL_x = 50.9151 ± 0.001 | UCL: 50.9151 ± 0.001 | **PASS** |
| TC-SPC-XBR-015 | LCL_x = 49.4149 ± 0.001 | LCL: 49.4149 ± 0.001 | **PASS** |
| TC-SPC-XBR-016 | UCL_R = 2.7482 ± 0.001 | UCL_R: 2.7482 ± 0.001 | **PASS** |

## 6.5 jrc_spc_xbar_s — 15/15 PASS

Script: jrc_spc_xbar_s.R — X-bar & S (X-bar and Standard Deviation) chart. Test data: xbar_s_stable.csv (stable), xbar_s_ooc.csv (out-of-control).

| TC ID | Description | Expected | Result |
|---|---|---|---|
| TC-SPC-XBS-001 | Stable dataset — exit 0, X-bar and S sections and verdict present | Exit 0, X-bar, Std Dev sections, verdict | **PASS** |
| TC-SPC-XBS-002 | Stable data — IN CONTROL verdict | Exit 0, 'IN CONTROL' or 'STABLE' | **PASS** |
| TC-SPC-XBS-003 | OOC dataset — exit 0, OUT OF CONTROL | Exit 0, 'OUT OF CONTROL' or 'SIGNALS' | **PASS** |
| TC-SPC-XBS-004 | OOC dataset — subgroup sg09 flagged as violation | Exit 0, 'sg09' in output | **PASS** |
| TC-SPC-XBS-005 | --ucl / --lcl flags accepted | Exit 0, user-specified UCL mentioned | **PASS** |
| TC-SPC-XBS-006 | PNG written to ~/Downloads/ | PNG present | **PASS** |
| TC-SPC-XBS-007 | No arguments — non-zero exit, usage message | Exit ≠ 0, 'Usage' | **PASS** |
| TC-SPC-XBS-008 | File not found — non-zero exit | Exit ≠ 0 | **PASS** |
| TC-SPC-XBS-009 | Missing column — non-zero exit | Exit ≠ 0 | **PASS** |
| TC-SPC-XBS-010 | Unbalanced subgroups — non-zero exit | Exit ≠ 0 | **PASS** |
| TC-SPC-XBS-011 | Bypass protection | Exit ≠ 0, 'RENV_PATHS_ROOT' | **PASS** |
| TC-SPC-XBS-012 | Grand X-bar = 100.060 ± 0.001 (xbar_s_stable.csv) | Grand X-bar: 100.060 ± 0.001 | **PASS** |
| TC-SPC-XBS-013 | UCL_x = 100.5807 ± 0.001 | UCL: 100.5807 ± 0.001 | **PASS** |
| TC-SPC-XBS-014 | LCL_x = 99.5393 ± 0.001 | LCL: 99.5393 ± 0.001 | **PASS** |
| TC-SPC-XBS-015 | UCL_s = 0.8598 ± 0.001 | UCL_s: 0.8598 ± 0.001 | **PASS** |

## 6.6 jrc_spc_p — 13/13 PASS

Script: jrc_spc_p.R — P-chart (proportion nonconforming). Test data: p_stable.csv (stable), p_ooc.csv (out-of-control).

| TC ID | Description | Expected | Result |
|---|---|---|---|
| TC-SPC-P-001 | Stable dataset — exit 0, p-bar, control limits, verdict present | Exit 0, p-bar, UCL, verdict | **PASS** |
| TC-SPC-P-002 | Stable data — IN CONTROL verdict | Exit 0, 'IN CONTROL' or 'STABLE' | **PASS** |
| TC-SPC-P-003 | Known p_bar ≈ 0.041 in output | Exit 0, value near 0.041 | **PASS** |
| TC-SPC-P-004 | OOC dataset — exit 0, OUT OF CONTROL | Exit 0, 'OUT OF CONTROL' or 'SIGNALS' | **PASS** |
| TC-SPC-P-005 | OOC dataset — subgroup 13 flagged as violation | Exit 0, '13' in output | **PASS** |
| TC-SPC-P-006 | PNG written to ~/Downloads/ | PNG present | **PASS** |
| TC-SPC-P-007 | No arguments — non-zero exit, usage message | Exit ≠ 0, 'Usage' | **PASS** |
| TC-SPC-P-008 | File not found — non-zero exit | Exit ≠ 0 | **PASS** |
| TC-SPC-P-009 | Missing column — non-zero exit | Exit ≠ 0 | **PASS** |
| TC-SPC-P-010 | defectives > n — non-zero exit | Exit ≠ 0 | **PASS** |
| TC-SPC-P-011 | Bypass protection | Exit ≠ 0, 'RENV_PATHS_ROOT' | **PASS** |
| TC-SPC-P-012 | p-bar = 0.04160 ± 0.0001 (p_stable.csv) | p-bar: 0.04160 ± 0.0001 | **PASS** |
| TC-SPC-P-013 | UCL = 0.10150 ± 0.0010 (p_stable.csv) | UCL: 0.10150 ± 0.001 | **PASS** |

## 6.7 jrc_spc_c — 12/12 PASS

Script: jrc_spc_c.R — C-chart (count of defects per unit). Test data: c_stable.csv (stable), c_ooc.csv (out-of-control).

| TC ID | Description | Expected | Result |
|---|---|---|---|
| TC-SPC-C-001 | Stable dataset — exit 0, c-bar, control limits, verdict present | Exit 0, c-bar, UCL, verdict | **PASS** |
| TC-SPC-C-002 | Stable data — IN CONTROL verdict | Exit 0, 'IN CONTROL' or 'STABLE' | **PASS** |
| TC-SPC-C-003 | Known c_bar ≈ 4.960 in output | Exit 0, value near 4.96 | **PASS** |
| TC-SPC-C-004 | OOC dataset — exit 0, OUT OF CONTROL | Exit 0, 'OUT OF CONTROL' or 'SIGNALS' | **PASS** |
| TC-SPC-C-005 | OOC dataset — subgroup 12 flagged as violation | Exit 0, '12' in output | **PASS** |
| TC-SPC-C-006 | PNG written to ~/Downloads/ | PNG present | **PASS** |
| TC-SPC-C-007 | No arguments — non-zero exit, usage message | Exit ≠ 0, 'Usage' | **PASS** |
| TC-SPC-C-008 | File not found — non-zero exit | Exit ≠ 0 | **PASS** |
| TC-SPC-C-009 | Missing column — non-zero exit | Exit ≠ 0 | **PASS** |
| TC-SPC-C-010 | Bypass protection | Exit ≠ 0, 'RENV_PATHS_ROOT' | **PASS** |
| TC-SPC-C-011 | c-bar = 4.960 ± 0.001 (c_stable.csv) | c-bar: 4.960 ± 0.001 | **PASS** |
| TC-SPC-C-012 | UCL = 11.641 ± 0.010 (c_stable.csv) | UCL: 11.641 ± 0.010 | **PASS** |

# 7. Deviations

No deviations were identified during OQ execution. All 71 test cases passed.

## 8. Requirements Traceability Matrix

The following table maps each User Requirement (UR) to the test cases that verify it. All 27 URs are fully covered by the 71 test cases executed in this OQ.

| UR ID | Requirement Description | Verified By |
|---|---|---|
| UR-SPC-001 | I-MR chart computes X-bar, MR-bar, UCL_I, LCL_I, UCL_MR | TC-SPC-IMR-001..004, 012..015 |
| UR-SPC-002 | I-MR user-specified limits accepted | TC-SPC-IMR-005 |
| UR-SPC-003 | I-MR chart PNG to ~/Downloads/ | TC-SPC-IMR-006 |
| UR-SPC-004 | I-MR error handling (no args, file not found, missing col, too few obs) | TC-SPC-IMR-007..010 |
| UR-SPC-005 | I-MR bypass protection | TC-SPC-IMR-011 |
| UR-SPC-006 | X-bar & R computes grand X-bar, R-bar, UCL_x, LCL_x, UCL_R | TC-SPC-XBR-001..004, 013..016 |
| UR-SPC-007 | X-bar & R user-specified limits | TC-SPC-XBR-005 |
| UR-SPC-008 | X-bar & R PNG to ~/Downloads/ | TC-SPC-XBR-006 |
| UR-SPC-009 | X-bar & R error handling (no args, file not found, missing col, unbalanced, size > 10) | TC-SPC-XBR-007..011 |
| UR-SPC-010 | X-bar & R bypass protection | TC-SPC-XBR-012 |
| UR-SPC-011 | X-bar & S computes grand X-bar, s-bar, UCL_x, LCL_x, UCL_s | TC-SPC-XBS-001..004, 012..015 |
| UR-SPC-012 | X-bar & S user-specified limits | TC-SPC-XBS-005 |
| UR-SPC-013 | X-bar & S PNG to ~/Downloads/ | TC-SPC-XBS-006 |
| UR-SPC-014 | X-bar & S error handling (no args, file not found, missing col, unbalanced) | TC-SPC-XBS-007..010 |
| UR-SPC-015 | X-bar & S bypass protection | TC-SPC-XBS-011 |
| UR-SPC-016 | P-chart computes p-bar, UCL, LCL | TC-SPC-P-001..005, 012..013 |
| UR-SPC-017 | P-chart PNG to ~/Downloads/ | TC-SPC-P-006 |
| UR-SPC-018 | P-chart error handling (no args, file not found, missing col, defectives > n) | TC-SPC-P-007..010 |
| UR-SPC-019 | P-chart bypass protection | TC-SPC-P-011 |
| UR-SPC-020 | C-chart computes c-bar, UCL, LCL | TC-SPC-C-001..005, 011..012 |
| UR-SPC-021 | C-chart PNG to ~/Downloads/ | TC-SPC-C-006 |
| UR-SPC-022 | C-chart error handling (no args, file not found, missing col) | TC-SPC-C-007..009 |
| UR-SPC-023 | C-chart bypass protection | TC-SPC-C-010 |
| UR-SPC-024 | All charts — OOC observations flagged with rule number | TC-SPC-IMR-004, XBR-004, XBS-004, P-005, C-005 |
| UR-SPC-025 | All charts — stable data produces IN CONTROL verdict | TC-SPC-IMR-002, XBR-002, XBS-002, P-002, C-002 |
| UR-SPC-026 | I-MR numeric correctness (Shewhart constants d2=1.128, D4=3.267) | TC-SPC-IMR-012..015 |
| UR-SPC-027 | X-bar & R/S, P, C numeric correctness | TC-SPC-XBR-013..016, XBS-012..015, P-012..013, C-011..012 |

## 9. OQ Conclusion

The Operational Qualification of the JR Validated Environment Statistical Process Control Module v1.0 is complete.

**Result: PASS**

All 71 automated test cases specified in JR-VP-SPC-001 v1.0 were executed on 2026-03-31 and 71 passed. Results are derived from the OQ evidence file listed below.

- jrc_spc_imr — I-MR (Individuals & Moving Range) Chart

- jrc_spc_xbar_r — X-bar & R (X-bar and Range) Chart
- jrc_spc_xbar_s — X-bar & S (X-bar and Standard Deviation) Chart
- jrc_spc_p — P-Chart (Proportion Nonconforming)
- jrc_spc_c — C-Chart (Count of Defects per Unit)

*Evidence of test execution is archived at: spc_oq_execution_20260331T194057.txt*

The Statistical Process Control module is released as version 1.0 of repos/spc/. Any subsequent change to the scripts, wrappers, or help files within repos/spc/ will require a change assessment and, if the change affects functionality covered by this OQ, re-execution of the affected test cases.